HSLU
Hochschule Luzern
Departement Informatik

Project Proposal
MLOPS H26

Proposal for Machine Learning Operations Project

Semester: HS26

Project Title	Swiss Control Area Balance Forecast
Course / Module	MLOPS H26
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Public Repository	GitHub Swiss Control Area Balance Forecast

1 Short Project Description

The project forecasts the imbalance of the Swiss control area (Ausgleichsenergie) using data published daily by Swissgrid, retraining the model every day and serving the latest forecast on a small dashboard.

2 Problem Statement

Problem: A modern problem of electric grids is no longer the production, which has countless options to be sustained, but rather system imbalance. Internal photovoltaic energy production spikes during the day and needs to be sustained at night and during winter. On the external side, neighbouring countries have their own energy overflow, which needs to be contained within an acceptable range to maintain net stability. Energy companies have their internal observations, but normal people do not have a way to predict net stability.

Prediction: Swiss control area imbalance (Total System Imbalance, MW)

Horizon: Every night at 23:20 UTC on day D, one forecast for all 96 quarter-hours of D+1 (15-min resolution, non-recursive). Only values published before the issue time are used.

Scope: Swiss control area, as published by Swissgrid. In detail, this means one single control area for all of Switzerland, without subdivisions for cantons.

Success criterion: MAE on the 24h-ahead imbalance forecast, computed as $\frac{1}{N}\sum|\hat{y}_t - y_t|$ over 15-min steps.

Baseline: persistence with a lag that exists at issue time, with its MAE computed directly from historical data before any model training. The one-day lag (t-96) is not known for the last steps of D+1, so the baselines are the same quarter-hour two days earlier (t-192, 96.09 MW) and seven days earlier (t-672, 94.26 MW), measured on 11,187 test rows from 01.07.2026.

Target: MAE below 80.1 MW, which is 15% better than the best baseline (t-672, 94.26 MW).

Evaluation: The original forecast is the one that is evaluated. A recalculation for a past period is labelled separately and never replaces it. The live MAE is scored against first-published values (the daily snapshot); a second figure against corrected values is reported after the revision window. Each forecast is stored with its creation time and model version.

3 Originality and Motivation

The motivation stems from my work, since we often have to work with energy data, and net stability is a major concern for our customers. Additionally, most of the energy-inspired projects covered on

mlops-lab.ch or in the KTH ID2223 primarily focus on PV, which makes this subject doubly suitable.

4 Data sources and features

Source: Swissgrid "Control Area Balance" timeline. The daily file holds a rolling window of about 24 hours and is updated every 15 minutes with a delay of at most 10 minutes.

History: Yearly CSVs from 01.11.2024, about 69.4k quarter-hours, growing by 96 per day. Used for initial training and to fill out missing values.

Access: The download links are read from the Swissgrid energy data page. The data is a non-binding offer without a guarantee of correctness or availability.

Fallback: ENTSO-E imbalance data, which offers only hourly values per country. It is used only if the Swissgrid data is not updated.

Data quality: Values are only reliable after 6 months, since revisions are possible due to Swissgrid suppliers correcting existing issues. Every download is kept as a dated copy, so the first-published values can be reconstructed.

Label: Total System Imbalance (MW), independently measured by Swissgrid

Features: Imbalance and balancing energy price at t-192 and t-672 only, since shorter lags are not known at issue time and the price is set from the imbalance of the same quarter-hour. Calendar features (quarter of day, day of week, month) in Zurich time.

Leakage: Out-of-time split with the test period starting on 01.07.2026. A test checks that no feature uses a value published after the issue time.

5 System Design

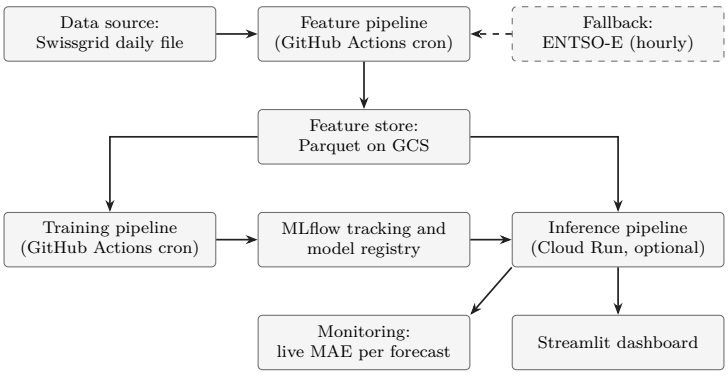


Figure 1: FTI pipeline with data source, model registry and monitoring.

Feature Store: Parquet files on GCS, one dated copy per run. PostgreSQL is used locally during development. Cloud SQL is optional and not part of the core, since a standing instance adds more cost and operation than needed.

Experiment Tracking: MLflow tracks daily retraining runs against the baseline and holds the model registry. Model artefacts are stored on GCS. Chosen because the exercises introduced a basic setup.

Orchestration: GitHub Actions cron runs the feature and training pipelines daily. Chosen since it needs no standing infrastructure.

Serving: A Cloud Run Service inference pipeline (optional) is called from a Streamlit dashboard to serve prognosis data on demand. It will serve as the required public-facing frontend.

Monitoring: The live MAE is computed per forecast from the stored forecast table.

Stretch (optional): MeteoSwiss solar/precipitation features to test weather's effect on imbalance (planned for M3, no weather data is ingested in M2). Drift monitoring on the daily retrain.